

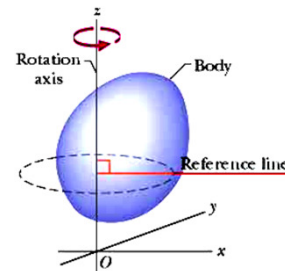


Chapter 10 – Rotation

Angular position, velocity and acceleration
 Rigid objects – constant angular acceleration
 Angular and linear variables Torque
 Moment of inertia Rotational KE
 Energy in Rotation

Rotational Variables

- ▶ Rotation of a **rigid body**
- ▶ About a fixed **rotation axis**.
- ▶ **Rigid Body** – an object that can rotate with all its parts locked together and without any change in its shape.
- ▶ **Fixed axis** – Rotation occurs about an axis that does not move.
- ▶ **Every point** of the body moves in a **circle**
- ▶ whose **center lies on the rotation axis**,
- ▶ and moves through the **same angle** during a **time interval**.



Rotational Variables

Angular position:

$$\theta = \frac{s}{r}$$

Angular displacement:

$$\Delta\theta = \theta_f - \theta_i$$

Average angular velocity:

$$\omega_{avg} = \frac{\theta_f - \theta_i}{t_f - t_i} = \frac{\Delta\theta}{\Delta t}$$

Instant. Angular velocity:

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$$

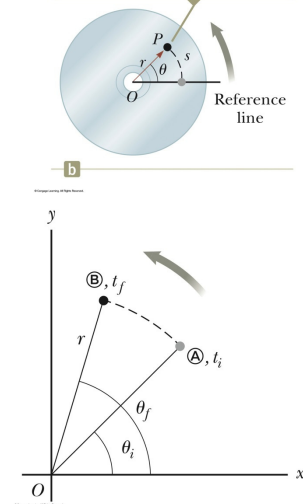
Average angular acceleration:

$$\alpha_{avg} = \frac{\omega_f - \omega_i}{t_f - t_i} = \frac{\Delta\omega}{\Delta t}$$

Instantaneous angular acceleration:

$$\alpha = \frac{d\omega}{dt}$$

As the disc rotates, a particle at P moves through an arc length s on a circular path of radius r . The angular position of P is θ .



3

Rotational Variables

- ▶ Reference line $\perp$ to rotation axis

- ▶ **Rule:** An angular displacement in the counter-clockwise direction is positive, and one in the clockwise direction is negative.

- ▶ θ is the angle in radians

$$\theta(\text{rad}) = \frac{2\pi}{360^\circ} \theta(\text{deg})$$

- ▶ The displacement **does not reset** θ to 0 after each rotation.

4

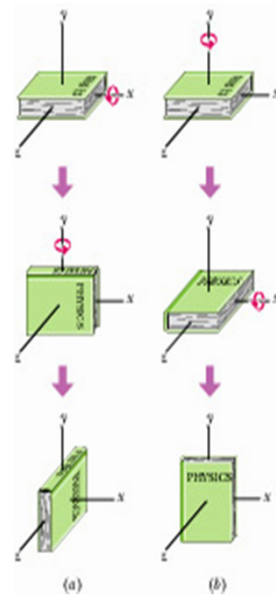
Example

- ▶ What is the angular speed of (a) second hand, (b) minute hand, and (c) hour hand of a smoothly running analogue watch? Give your answer in rad/s

5

Vectors?

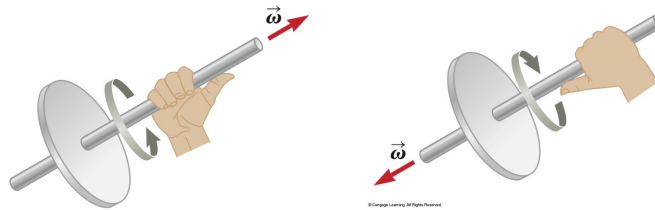
- ▶ Position, velocity and acceleration of a single particle – described by vectors. +ve and -ve signs are used to indicate direction.
- ▶ Rigid bodies can rotate about a fixed axis – either clockwise or counter-clockwise (- or + signs are used again).
- ▶ **Angular velocity ω** and the **angular acceleration α** can be treated as **vectors**, but **angular displacement** is **not a vector** quantity.



6

Rotational Variables – Vectors?

- ▶ **RH Rule:** Curl the fingers of RH in direction of rotation. Your thumb points along the direction of the angular velocity vector.



7

Constant Angular Velocity

- ▶ $\omega = \frac{d\theta}{dt}$

$$d\theta = \omega dt$$

$$\int_{\theta_0}^{\theta} d\theta = \omega \int_{t=0}^t dt$$

$$[\theta]_{\theta_0}^{\theta} = \omega [t]_0^t$$

- ▶ $\theta - \theta_0 = \omega t$

- ▶ For 1 revolution, $t = T$ and $\theta = 2\pi$

$$\therefore 2\pi = 0 + \omega t \quad T = \frac{2\pi}{\omega} = \text{period}$$

$$f = \frac{1}{T} = \frac{\omega}{2\pi} = \text{frequency}$$

8

Constant Angular Acceleration

- Derivation of these equations for this special case is the same as those for linear motion with constant acceleration.

Translational Motion Rotational Motion

$$x \leftrightarrow \theta$$

$$v \leftrightarrow \omega$$

$$a \leftrightarrow \alpha$$

$$v = v_0 + at \leftrightarrow \omega = \omega_0 + \alpha t \quad (\text{eq. 1})$$

$$x = x_0 + v_0 t + \frac{1}{2}at^2 \leftrightarrow \theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2 \quad (\text{eq. 2})$$

$$v^2 - v_0^2 = 2a(x - x_0) \leftrightarrow \omega^2 - \omega_0^2 = 2\alpha(\theta - \theta_0) \quad (\text{eq. 3})$$

$$x = x_0 + \frac{1}{2}(v + v_0)t \leftrightarrow \theta = \theta_0 + \frac{1}{2}(\omega + \omega_0)t \quad (\text{eq. 4})$$

$$x = x_0 + v_0 t - \frac{1}{2}at^2 \leftrightarrow \theta = \theta_0 + \omega_0 t - \frac{1}{2}\alpha t^2 \quad (\text{eq. 5})$$

9

Linear and Angular Variables.

Position: $s = \theta r$

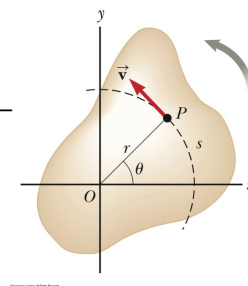
Speed: $\frac{ds}{dt} = \frac{d\theta}{dt} r$
 $v = \omega r$

- Translational velocity – tangential velocity.
- Angular velocity

Period: $\frac{2\pi r}{v} = \frac{2\pi}{\omega}$

Acceleration: $\frac{dv}{dt} = \frac{d\omega}{dt} r$
 $a_t = \alpha r$
 $a_r = \frac{v^2}{r} = \omega^2 r$

- Translational acceleration – tangential acceleration.
- Angular acceleration
- Radial acceleration

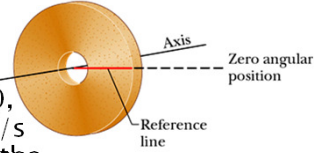


Radians!!

10

Example

- ▶ A grindstone rotates at constant angular acceleration $\alpha = 0.35 \text{ rad/s}^2$. At time $t = 0$, it has an angular velocity of $\omega_0 = -4.6 \text{ rad/s}$ and a reference line on it is horizontal, at the angular position $\theta_0 = 0$.



- At what time after $t = 0$ is the reference line at the angular position $\theta = 5 \text{ rev}$?
- Describe the grindstone's rotation between $t = 0$ and $t = 32 \text{ s}$.
- At what time t does the grindstone momentarily stop?

11

Example

The angular position of a point on a rotating wheel is given by $\theta = 2.0 + 4.0t^2 + 2.0t^3$, where θ is in radians and t is in seconds. At $t = 0$, what are (a) the points angular position and (b) its angular velocity? (c) What is its angular velocity at $t = 4.0 \text{ s}$? (d) Calculate its angular acceleration at $t = 2.0$. (e) Is its angular acceleration constant?

12

Example

A flywheel turns through 40 rev as it slows from an angular speed of 1.5 rad/s to a stop.

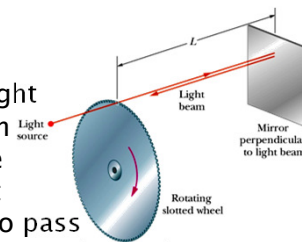
- (a) Assuming a constant angular acceleration, find the time for it to come to rest.
- (b) What is its angular acceleration?
- (c) How much time is required for it to complete the first 20 of the 40 revolutions?

- $\Delta\theta = 40 \text{ rev} = 80\pi$
- $\omega_0 = 1.5 \text{ rad/s}$

13

Example

An early method of measuring the speed of light makes use of a rotating slotted wheel. A beam of light passes through one of the slots at the outside edge of the wheel, travels to a distant mirror, and returns to the wheel just in time to pass through the next slot in the wheel. One such slotted wheel has a radius of 5.0 cm, and 500 slots around its edge. Measurements taken when the mirror is $L = 500\text{m}$ from the wheel indicated a speed of light $3 \times 10^5 \text{ km/s}$ (a) What is the constant angular speed of the wheel? (b) What is the linear speed of a point on the edge of the wheel?



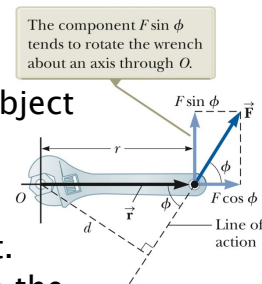
Torque

- ▶ Opening a heavy door
 - Where is door knob positioned?
 - Where do you normally apply the force?
 - What direction do you apply the force?
 - If a force is not applied at the edge of the door and at 90° to the door it is more difficult to move the door.

15

Torque

- ▶ The tendency of a force to rotate an object about some axis is measured by the torque $\vec{\tau}$.
- ▶ Consider the wrench which rotates around an axis passing through a bolt.
- ▶ The force $\vec{F}$ is applied at an angle ϕ to the horizontal direction at a distance $\vec{r}$.
- ▶ The components of the force:
 - radial component - no rotation $F_r = F \cos \phi$
 - tangential component - rotation $F_t = F \sin \phi$
- ▶ The ability of a force to rotate an object depends on the **magnitude** of the force and **position** at which it is applied.



Torque: $\tau = (r)(F \sin \phi) = (r \sin \phi)(F)$

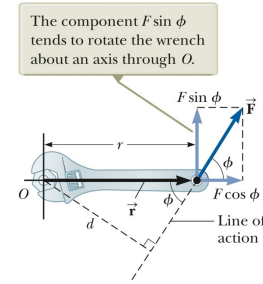
16

Torque

$$\tau = (r)(F \sin \phi) = rF_{\perp}$$

$$\tau = (r \sin \phi)(F) = r_{\perp} F$$

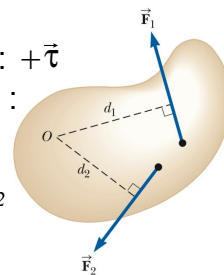
- ▶ $r \sin \phi = d$: moment arm
- ▶ Torque is the magnitude of force times the perpendicular distance from the rotation axis to the line of action of the force.
- ▶ Torque is a vector and can also be written as a cross product: $\vec{\tau} = \vec{r} \times \vec{F}$
- ▶ **RH Rule** – move $\vec{F}$ to O , curl fingers from $\vec{r}$ to $\vec{F}$, thumb shown direction of $\vec{\tau}$.



17

Torque

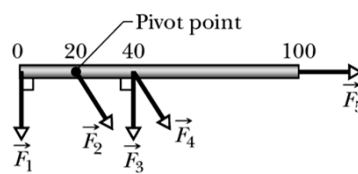
- ▶ If there are two or more forces acting on a rigid object, each force tends to produce rotation about the axis through O .
- ▶ $\vec{F}_1$ tends to cause anticlockwise rotation: $+\vec{\tau}$
- ▶ $\vec{F}_2$ tends to cause anticlockwise rotation : $-\vec{\tau}$
- ▶ $\sum \tau = \tau_1 + \tau_2 = F_1 r_1 \sin \phi_1 - F_2 r_2 \sin \phi_2 = F_1 d_1 - F_2 d_2$
- ▶ Units: N.m
- ▶ This is not the same as work, although they have the same units.



18

Example

- ▶ The Figure shows an overhead view of a meter stick that can pivot about the dot at the position marked 20 (for 20 cm). All five forces on the stick are horizontal and have the same magnitude. Rank the forces according to the magnitude of the torque they produce, greatest first.

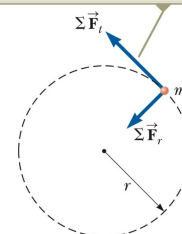


19

Rigid Object under a net force

- ▶ Previously – Net force results in acceleration of an object in a straight line.
- ▶ Consider a particle with mass m that is moving in a circle with radius r .
- ▶ Particle is under the influence of a net tangential force and a net radial force.
- ▶ The radial force causes circular motion with centripetal acceleration.
- ▶ Tangential force causes a tangential acceleration.

The tangential force on the particle results in a torque on the particle about an axis through the center of the circle.



20

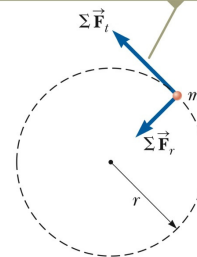
Rigid Object under a net force

- ▶ The radial force causes circular motion with centripetal acceleration.
- ▶ Tangential force causes a tangential acceleration.

$$\begin{aligned}\sum F_t &= ma_t \\ \sum \tau &= \sum F_t r = (ma_t) r \\ \sum \tau &= m(\alpha r) r = (mr^2) \alpha \\ I &= mr^2 \\ \sum \tau &= I \alpha\end{aligned}$$

- ▶ The same form as Newton's second law seen previously

The tangential force on the particle results in a torque on the particle about an axis through the center of the circle.



21

Rigid Object under a net force

- ▶ Consider an arbitrary rigid object that rotates about a fixed axis.
- ▶ We can consider the object to consist of a collection of small masses m_i .
- ▶ Choose a set of axes. Each particle moves in a circle around the origin with a tangential acceleration due to the force:

$$F_i = m_i a_i$$

$$\tau_i = F_i r_i = m_i a_i r_i$$

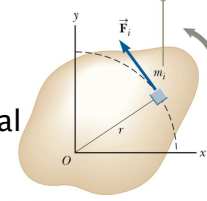
$$\tau_i = m_i r_i^2 \alpha$$

$$\sum \tau_{ext} = \sum \tau_i = \sum_i m_i r_i^2 \alpha = \left(\sum_i m_i r_i^2 \right) \alpha$$

$$I = \sum_i m_i r_i^2 \quad \text{Moment of inertia}$$

$$\sum \tau = I \alpha \quad I \text{ plays the same role here as mass plays in Newton 2}$$

The particle of mass m_i of the rigid object experiences a torque in the same way that the particle in Figure 10.10 does.



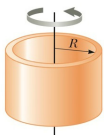
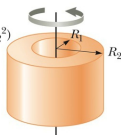
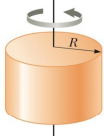
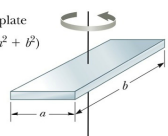
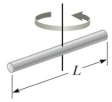
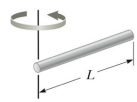
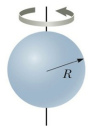
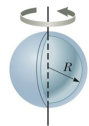
22

Moment of Inertia

- ▶ Moment of inertia, I , depends on the shape of the object
- ▶ I depends on the position of the rotation axis.
- ▶ There is a minimum value for I when the rotation axis passes through the COM.
- ▶ Table 10.2 gives the equations for I for various shapes with rotation axes passing through the COM.

23

Table 10.2 Moments of Inertia of Homogeneous Rigid Objects with Different Geometries

Hoop or thin cylindrical shell $I_{\text{CM}} = MR^2$ 	Hollow cylinder $I_{\text{CM}} = \frac{1}{2} M(R_1^2 + R_2^2)$ 
Solid cylinder or disk $I_{\text{CM}} = \frac{1}{2} MR^2$ 	Rectangular plate $I_{\text{CM}} = \frac{1}{12} M(a^2 + b^2)$ 
Long, thin rod with rotation axis through center $I_{\text{CM}} = \frac{1}{12} ML^2$ 	Long, thin rod with rotation axis through end $I = \frac{1}{3} ML^2$ 
Solid sphere $I_{\text{CM}} = \frac{2}{5} MR^2$ 	Thin spherical shell $I_{\text{CM}} = \frac{2}{3} MR^2$ 

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24

Moment of Inertia

- ▶ For a continuous mass: $I = \lim_{\Delta m \rightarrow 0} \sum_i r_i^2 \Delta m_i = \int r^2 dm$
- ▶ Then
$$\rho \equiv \frac{dm}{dV} \quad \therefore dm = \rho dV$$

$$\begin{aligned}
 I &= \int R^2 dm \\
 &= \int \rho R^2 dV && \text{-- 3-dimensions} \\
 &= \int \sigma r^2 dV && \text{-- 2-dimensions} \\
 &= \int \lambda x^2 dx && \text{-- 1-dimension}
 \end{aligned}$$

- ▶ ρ – Volume density
- σ – surface mass density
- λ – Linear mass density

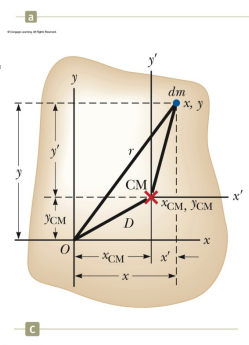
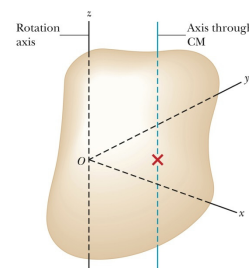
25

Parallel Axis Theorem

- ▶ To find the Rotational inertia I of a body of mass M about an axis which does not pass through the centre of mass.
- ▶ If I_{com} is known then: $I = I_{com} + MD^2$

where D is the distance from the axis to the axis through the centre of mass

- ▶ The **moment of inertia** of a body **about any axis** is equal to the **moment of inertia** ($= MDh^2$) it would have about that axis if all its **mass were concentrated at its center of mass** plus its **moment of inertia** ($= I_{com}$) about a **parallel axis through its center of mass**.



26

Rotational KE

- ▶ Consider the rotating rigid body
- ▶ Divide body into small parts of masses $m_1, m_2, m_3, \dots m_i$
- ▶ Kinetic energy of rotation

$$K = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \frac{1}{2}m_3v_3^2 + \dots + \frac{1}{2}m_iv_i^2$$

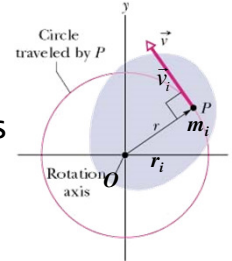
$$= \sum \frac{1}{2}m_iv_i^2 = \sum \frac{1}{2}m_i(\omega r_i)^2 = \frac{1}{2}\left(\sum m_ir_i^2\right)\omega^2$$

- ▶ **Rotational Inertia** (moment of Inertia) -:

$$I = \sum m_ir_i^2$$

- ▶ Rotational KE:

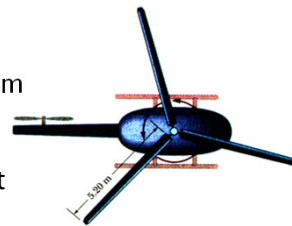
$$K = \frac{1}{2}I\omega^2$$



27

Example

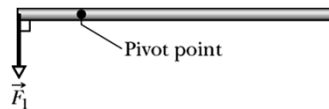
Each of the 3 helicopter rotor blades is 5.20 m long and has a mass of 240 kg. The rotor is rotating at 350 rev/min. (a) What is the rotational inertia of the rotor assembly about the axis of rotation? (Each blade can be considered to be a thin rod.) (b) What is the kinetic energy of rotation?



28

Example

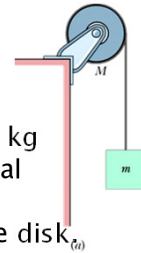
The figure shows an overhead view of a meter stick that can pivot about the point indicated which is to the left of the stick's midpoint. Two horizontal forces are applied to the stick. Only one is shown. The second force is perpendicular to the stick and is applied to the right end. If the stick is not to turn (a) what should the direction of the force be and (b) should this second force be greater or less than or equal to the first?



29

Example

- ▶ The figure shows a uniform disk with mass $M = 2.5$ kg and radius $R = 20$ cm, mounted on a fixed horizontal axle. A block with mass $m = 1.2$ kg hangs from a massless cord that is wrapped around the rim of the disk. Find the acceleration of the falling block, the angular acceleration of the disk and the tension in the cord. The cord does not slip, and there is no friction at the axle.



30

Energy and Rotational

▶ $\Delta K = \frac{1}{2} I \omega_f^2 - \frac{1}{2} I \omega_i^2 = W$

▶ $W = \int_{\theta_i}^{\theta_f} \tau d\theta$

▶ $P = \frac{dW}{dt} = \tau \frac{d\theta}{dt} = \tau \omega$

31

Example

- ▶ Attached to each end of a thin steel rod of length 1.20m and mass 6.40 kg is a small ball of mass 1.06 kg. The rod is constrained to rotate in a horizontal plane about a vertical axis through its midpoint. At a certain instant, it is observed to be rotating with an angular velocity of 39.0 rev/s. Because of friction, it comes to rest 32s later. Assuming constant frictional torque, compute (a) the angular acceleration, (b) the retarding torque exerted by the friction, (c) the total mechanical energy dissipated by the friction, and (d) the number of revs executed during the 32.0 s. (e) Now suppose that the frictional torque is known not to be constant, which, if any, of the quantities above can still be computed without additional information?

- $L = 1.20 \text{ m}$
- $M_{\text{rod}} = 6.40 \text{ kg}$
- $M_{\text{ball}} = 1.06 \text{ kg}$
- $\omega_0 = 39 \text{ rev/s} = 245 \text{ rad/s}$
- $\omega = 0 \text{ rad/s}$
- $t = 32 \text{ s}$

32



Chapter 11: Angular momentum

Vector product (Cross product)
Torque

Cross Products

- ▶ Review the rules regarding cross-products
- ▶ Review application of the right hand rule to determine the direction of the resultant vector of the cross-product of two vectors.
- ▶ Chapter 11

Torque

- ▶ Torque is defined as:

$$\tau = (r)(F \sin \phi) = (r \sin \phi)(F)$$

- Torque can be written as a vector cross product:

$$\vec{\tau} = \vec{r} \times \vec{F}$$

- **Right hand rule** - move $\vec{F}$ to O , curl fingers from $\vec{r}$ to $\vec{F}$, thumb shows direction of $\vec{\tau}$.
- Read section 11-6 for more information and Sample Problems

35

Problem 23

- ▶ Force $\vec{F} = (2.0 \text{ N})\hat{i} - (3.0 \text{ N})\hat{k}$ acts on a pebble with position vector $\vec{r} = (0.5 \text{ m})\hat{j} - (2.0 \text{ m})\hat{k}$ relative to the origin. In unit-vector notation, what is the resulting torque on the pebble about (a) the origin and (b) the point (2.0m, 0, -3.0 m)?

36